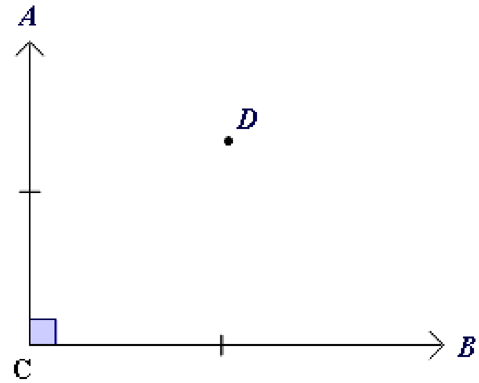
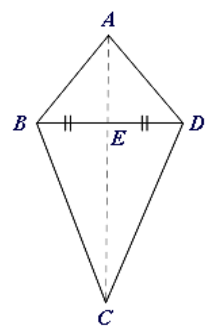
Midpoints and segment bisectors

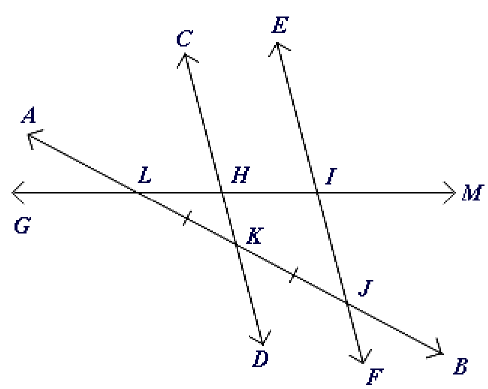
1 From the diagram:
Is C the midpoint of $\overline{AB}$?



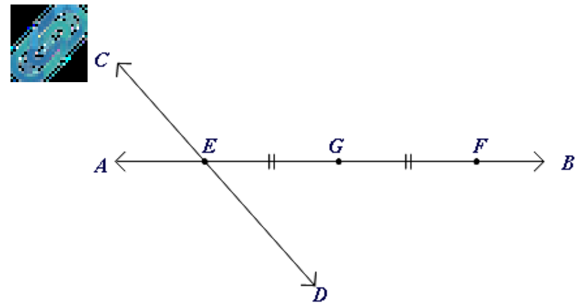
2 From the diagram:
What can be said about point E?



3 From the diagram:
Identify the midpoint of $\overline{LJ}$.

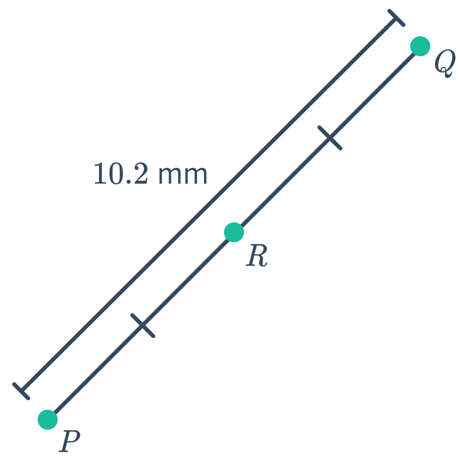


4 From the diagram:

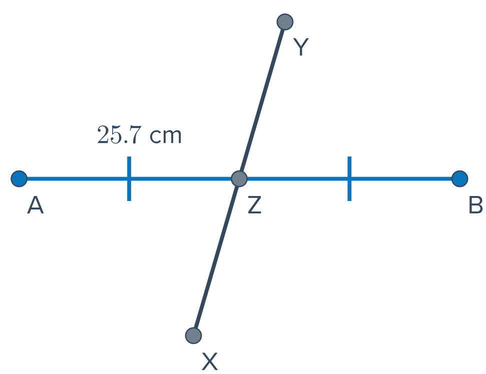


- **a** Identify the midpoint of $\overline{EF}$.
- **b** Which line segment has a length equivalent to $2 \times EG$?

5 Using the diagram shown, find PR.

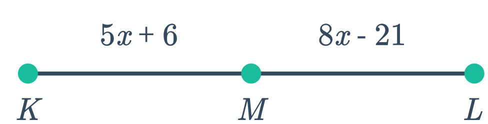


6 Using the diagram shown, find AB.

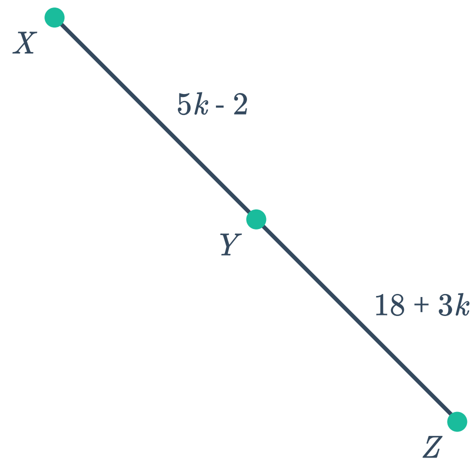


7 In the diagram shown, M is the midpoint of $\overline{KL}$.

- **a** Solve for the value of x.
- **b** Find ML.



8 In the diagram shown, Y is the midpoint of $\overline{XZ}$.



a Solve for the value of k.

b Find XZ.

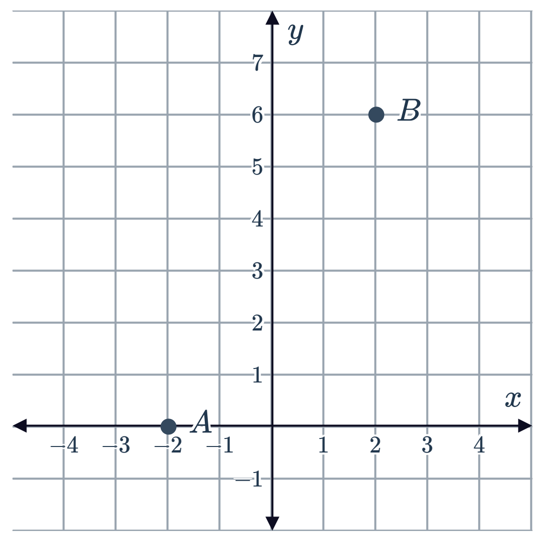
9 State the formula that describes the coordinates of the midpoint M of the points (x_1, y_1) and (x_2, y_2).

10 M is the midpoint of $A\,(-2, 0)$ and $B\,(2, 6)$.

a Find the x-coordinate of M.

b Find the y-coordinate of M.

c Plot the point M on the coordinate plane.



11 For each of the following, find the coordinates of the midpoint of $\overline{AB}$.

a $A\,(3, 1)$ and $B\,(-2, 4)$

b $A\,(-6, -5)$ and $B\,(-16, -11)$

c $A\left(\frac{13}{3}, -1\right)$ and $B\left(-1, \frac{6}{5}\right)$

d $A\left(\frac{5}{2}, -\frac{5}{2}\right)$ and $B\left(\frac{9}{2}, -\frac{9}{2}\right)$

12 Given that M is the midpoint of $\overline{AB}$, find the coordinates of A.

a $B(1,4)$ and $M(9,7)$

b $B(3,5)$ and $M(0,-6)$

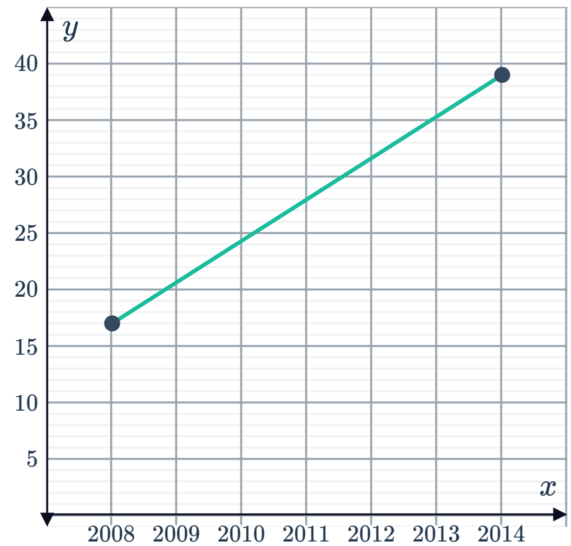
c $B(-7,2)$ and $M(1,-2)$

d $B(-9,2)$ and $M(1,-5)$

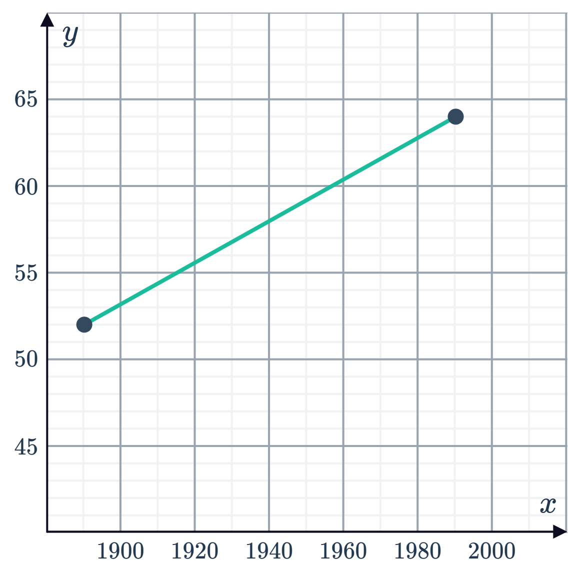
13 Find the midpoint of $A(-7m,-3n)$ and $B(-5m,-5n)$.

14 Lines of latitude and longitude measure position on the Earth's surface and work like coordinates. The first coordinate represents how far above or below the equator you are, and the second coordinate measures how far from Greenwich Mean Time you are.
A plane starts its flight at $(26^\circ\text{N}, 86^\circ\text{E})$. It is bound for its destination at $(24^\circ\text{N}, 108^\circ\text{E})$.
What will be its position halfway through the flight?

15 The graph shows the annual net profit (in millions) of a company over the last few years. It shows that its profit has been growing approximately linearly from $\$17$ million in 2008 to $\$39$ million in 2014.
By finding the midpoint of the line segment, determine the company's net profit in 2011.



16 The graph shows a straight line that approximates the global life expectancy of a child over a period of one hundred years.
The graph shows the life expectancy to be 52 years in 1890, and 64 years in 1990.
Use the midpoint formula to estimate the life expectancy of a child in 1940.



17 The table shows the number of smartphone users in a particular city.

Year	Number of smartphone users
2007	252 554
2011	334 084
2015	415 180

a Use the midpoint formula to estimate the number of smartphone users in 2009.

b Use the midpoint formula to estimate the number of smartphone users in 2013.

18 Q is the midpoint of $\overline{PR}$, with $PQ = 3x - 8$ and $QR = 2x + 3$.
Determine the length PQ.

19 $\overline{XY}$ bisects $\overline{UV}$ at W.
If $UW = 8x + 5$ metres, find an expression for the length UV.

20 $\overline{AB}$ bisects $\overline{CD}$ at E.
If $CE = 2\frac{1}{5}$ inches, determine the length CD.

21 The points P, Q, R, S, and T lie on a straight line such that $\overline{PQ}$, $\overline{QR}$, $\overline{RS}$, and $\overline{ST}$ are equidistant.
Given $P\,(-6, -4)$ and $T\,(-12, 16)$, find the coordinates of point:

a Q b R c S

22 $\overline{AB}$ has endpoints $A\left(3m - 8, \frac{n}{9}\right)$ and $B\,(-7m, 2n)$. The midpoint of the segment is at M.

a Form an expression for the x-coordinate of M.

b Form an expression for the y-coordinate of M.

23

a Find the midpoint of $A\,(-10, 10)$ and $B\,(16, 4)$.

b For what value of k will the line $y = 6x + k$ bisect the interval AB?

24 $M\,(4p + 1, 4q - 5)$ is the midpoint of $S\,(-14, -2)$ and $T\,(4, -8)$.

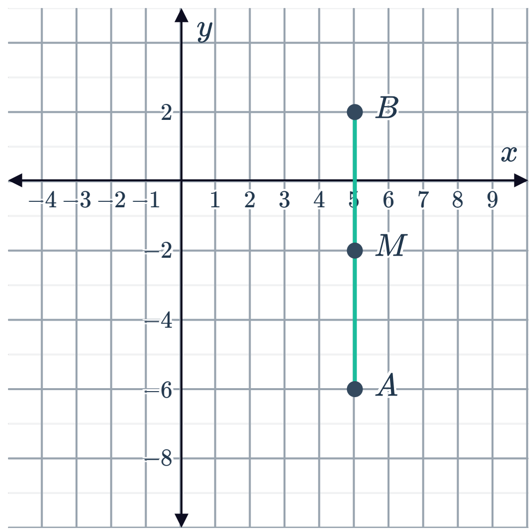
a Find the value of p. b Find the value of q.

Additional questions

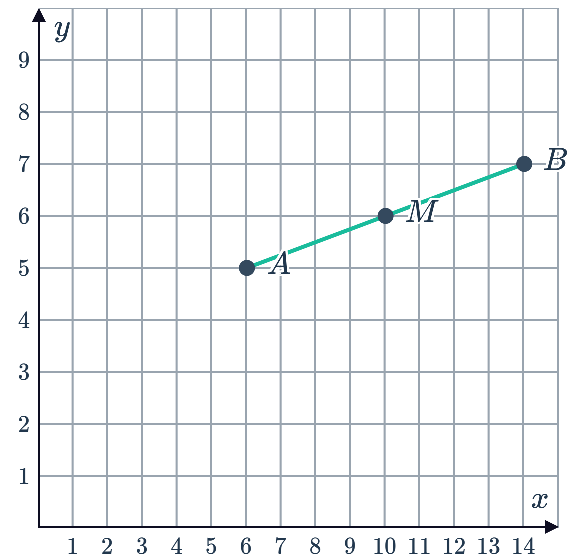
25 For the segment with endpoints, $A\left(x_1, y_1\right)$ and $B\left(x_2, y_2\right)$ state the formula for the midpoint, M, of $\overline{AB}$.

26 Consider the two points A and B with their midpoint M plotted on the coordinate plane. For each pair of coordinates, find their midpoint M:

a $A\left(5,-6\right)$ and $B\left(5,2\right)$

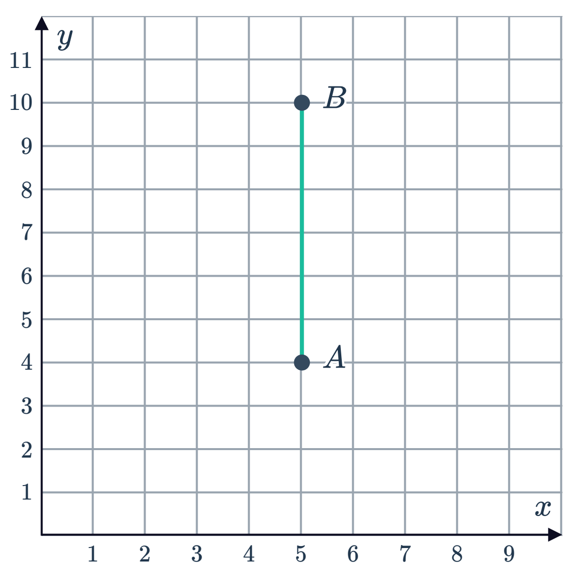


b $A\left(6,5\right)$ and $B\left(14,7\right)$

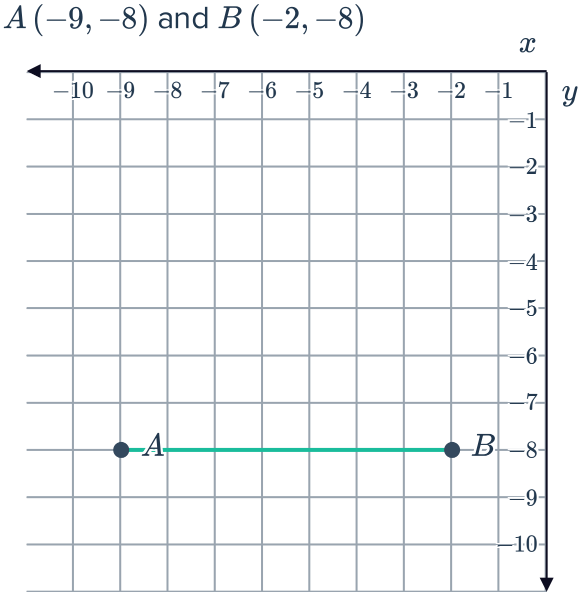


27 For each of the following, find the coordinates of M, the midpoint of $\overline{AB}$:

a $A\left(5,4\right)$ and $B\left(5,10\right)$

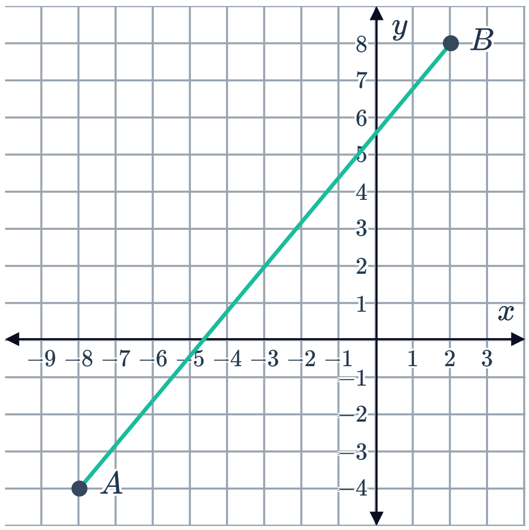


b $A\left(-9,-8\right)$ and $B\left(-2,-8\right)$

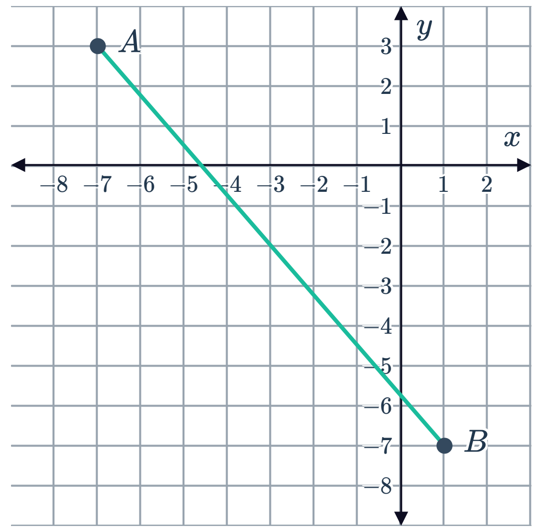


28 For each of the following find the coordinat[redacted] M, the midpoint of $\overline{AB}$:

a $A\,(-8,-4)$ and $B\,(2,8)$



b $A\,(-7,3)$ and $B\,(1,-7)$



29 For each pair of points A and B given:

i Find the coordinates of the midpoint M.

ii Plot $\overline{AB}$ and the point M on a coordinate plane to check your answer from part (a).

a $A\,(5,5)$ and $B\,(13,9)$

b $A\,(-9,1)$ and $B\,(5,-7)$

c $A\left(-\frac{3}{4},-6\right)$ and $B\left(\frac{11}{4},4\right)$

d $A\left(\frac{1}{2},-2\right)$ and $B\left(-2,\frac{7}{2}\right)$

30 Find the midpoint of $A\,(2m,5n)$ and $B\,(6m,n)$.

31 Given that M is the midpoint of A and B, find the coordinates of A in the following given points:

a $B\,(9,6)$ and $M\,(7,6)$

b $B\,(16,7)$ and $M\,(10,2)$

c $B\,(8,8)$ is $M\,(-2,0)$

d $B\,(2,-9)$ and $M\,(-1,-7)$

32 Lines of latitude and longitude measure position on the Earth's surface and work like coordinates. The first coordinate represents how far above or below the equator you are, and the second coordinate measures how far from Greenwich Mean Time you are.
A plane starts its flight at $(20^\circ\text{N}, 62^\circ\text{E})$. It is bound for its destination at $(52^\circ\text{N}, 146^\circ\text{E})$.
Assuming the plane flies directly to its destination, find its position a quarter of the way through the flight.

33 The points P, Q, R, S and T are collinear, an[redacted] $= QR = RS = ST$.
Find the points Q, R and S given $P\,(-4, -2)$ and $T\,(-12, 10)$.

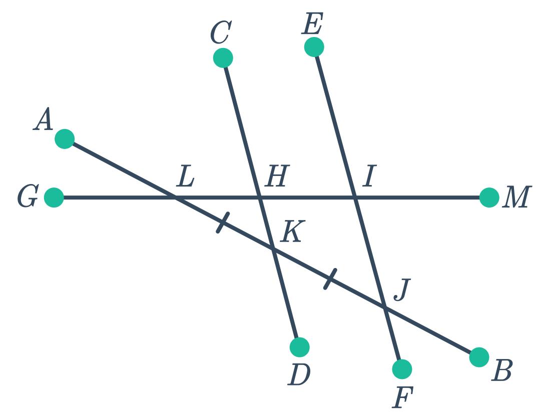
34 $M\,(4p + 2, 5q - 3)$ is the midpoint of $S\,(20, -12)$ and $T\,(-18, 6)$.

a Find the value of p.

b Find the value of q.

35 Use the diagram and the given information to answer each question:

- H is the midpoint of $\overline{LI}$
- $\overline{GM}$ bisects segment EJ



a List all pairs of congruent segments.

b Identify two additional segments that have been bisected, and the line which bisects them.

c Identify two additional midpoints.

36 Identify all midpoints in the diagram and the segments they bisect.